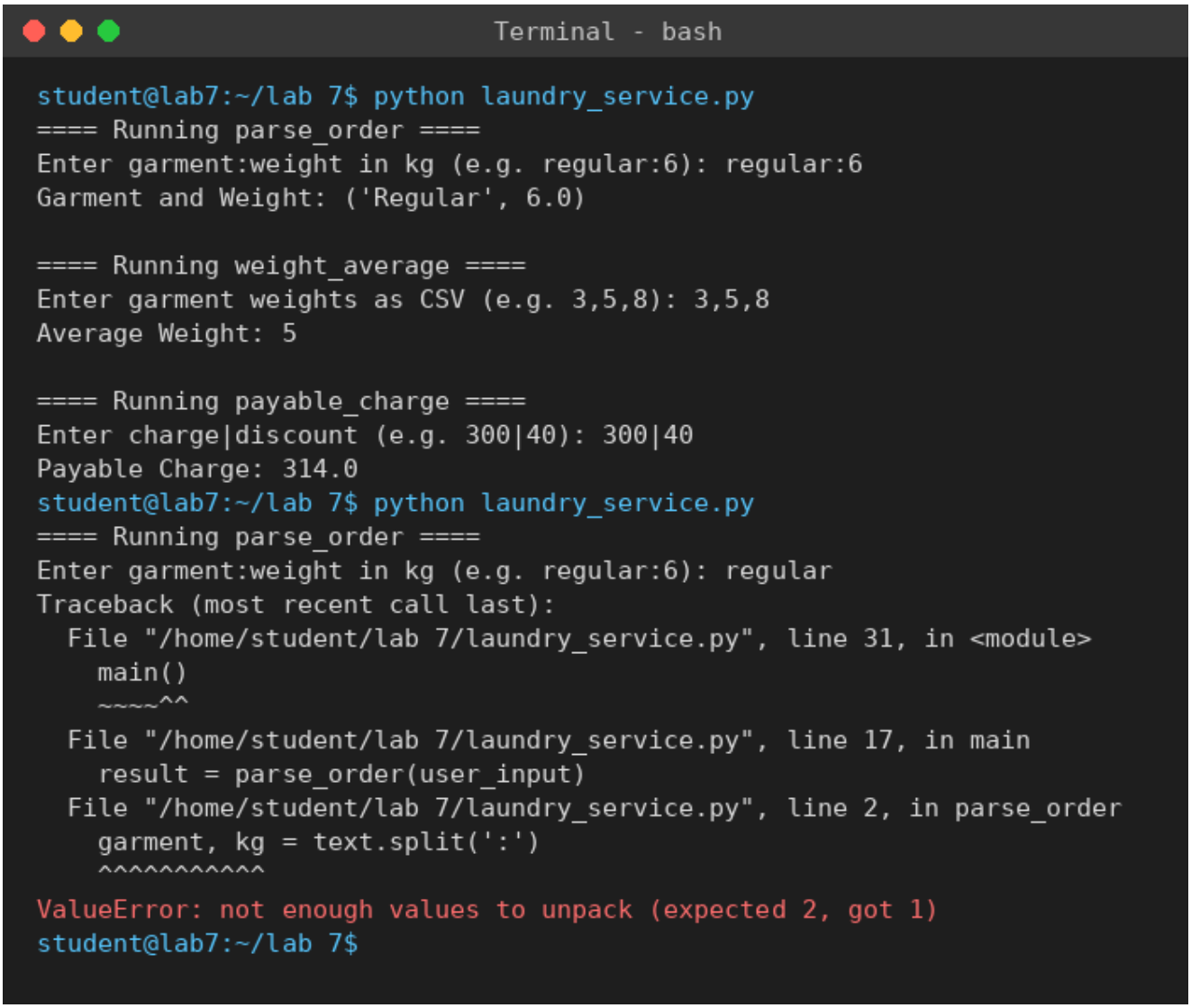
Lab 7: Fuzzing for Input Validation Bugs

Repository: L6_L7_41_laundry_service | Module: laundry_service.py | Tool: Hypothesis + pytest

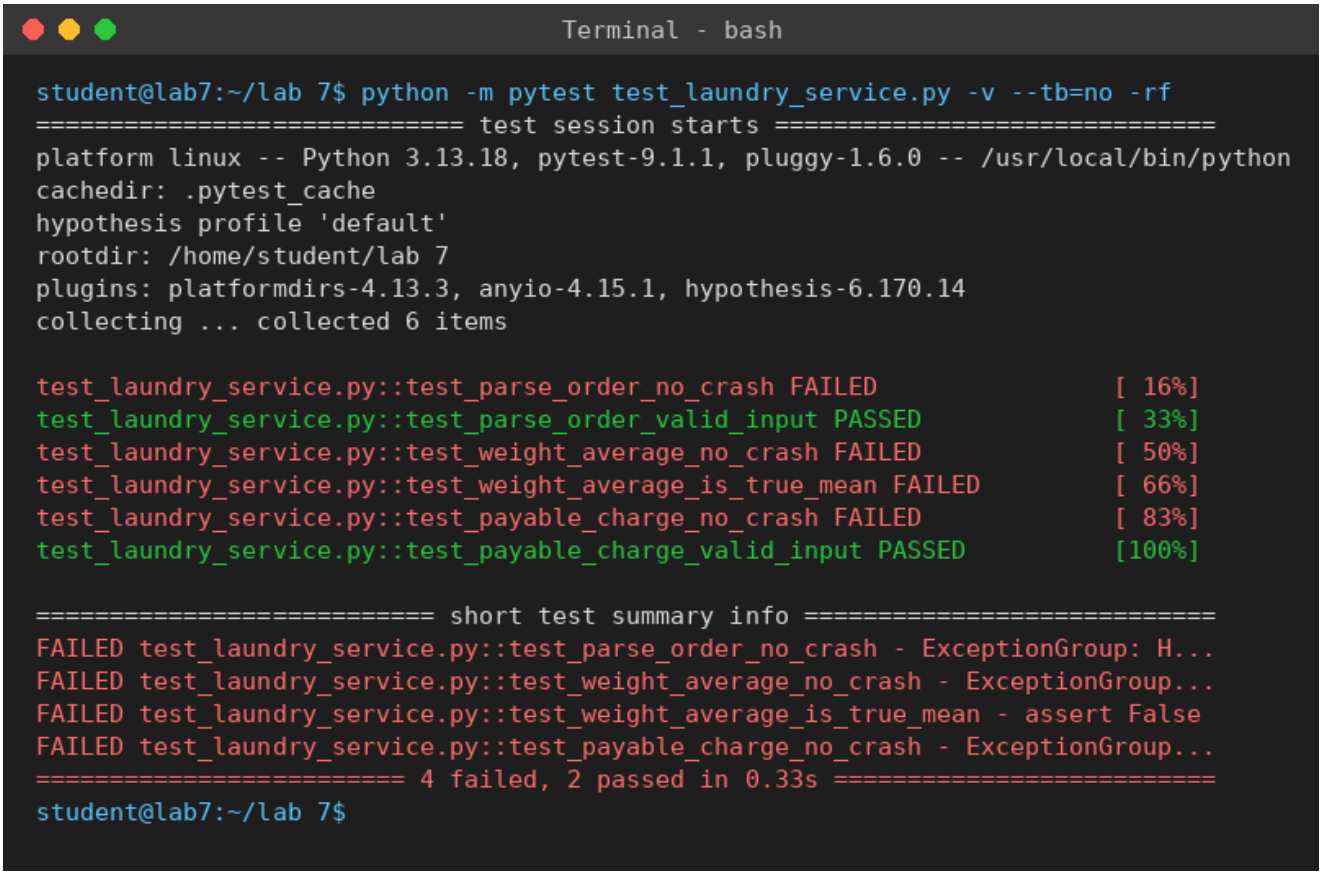
Step 2: Running the original (buggy) application

With the example inputs the program runs, but weight_average returns 5 for 3,5,8 instead of 5.33, because it uses integer division (//). With the input *regular* (no ':' separator) the program crashes with a ValueError.



Step 4: Hypothesis tests failing on the original code

Hypothesis generated random and malformed inputs. Four of the six properties fail: all three 'no crash' tests and the 'true mean' test for weight_average.



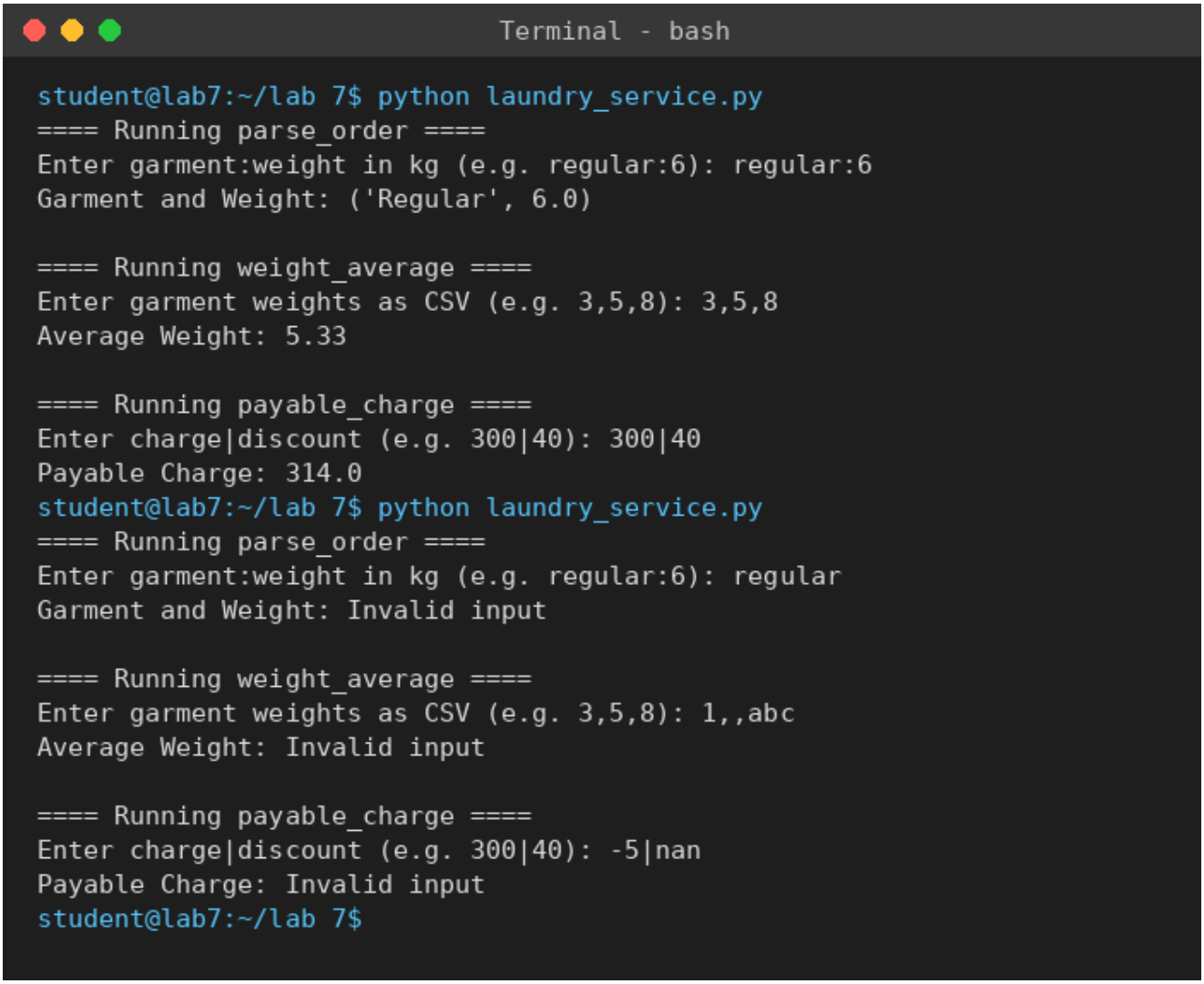
Bugs found by fuzzing

Function	Minimal failing input	Error / problem	Fix
parse_order	None	AttributeError (no .split)	Reject non-string input
parse_order	'' (no ':')	ValueError: not enough values to unpack	Require exactly one ':'
parse_order	':'	ValueError: could not convert '' to float	Validate number; reject empty name, nan/inf, kg <= 0
weight_average	None	AttributeError (no .split)	Reject non-string / empty input
weight_average	'' or '1,,3'	ValueError: invalid literal for int()	Reject empty / non-numeric items
weight_average	[0, 1] -> '0,1'	Returns 0 instead of 0.5 (// truncates)	Use true division, round to 2 dp
payable_charge	None	AttributeError (no .split)	Reject non-string input
payable_charge	'' (no '|')	ValueError: not enough values to unpack	Require exactly one '|'
payable_charge	'|'	ValueError: could not convert '' to float	Validate numbers; reject negatives, nan/inf, overflow

All three functions now return **None** for invalid input instead of crashing, and main() prints 'Invalid input' in that case.

Step 5: Running the fixed application

Valid inputs give the correct results (the average of 3,5,8 is now 5.33). Invalid inputs such as *regular*, *1,,abc* and *-5|nan* are handled without crashing.



Step 6: Hypothesis tests passing on the fixed code

All six property-based tests pass. Each no-crash test runs 500 generated inputs.

